
TRAVAUX DIRIG  S N   4 : Risk Convexification

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EXERCICE 1. We work in the framework of binary classification : we consider a random descriptor X with distribution P_X taking values in $\mathbb{R}^d$ ($d \in \mathbb{N}^*$), equipped with its Borel sigma-algebra, and a random label Y taking the value -1 or 1 .

The regression function (posterior probability) is denoted by $\eta : x \in \mathbb{R}^d \mapsto \mathbb{P}(Y = 1 \mid X = x) \in [0, 1]$. We assume that $0 < \eta(X) < 1$ almost surely.

Let $\mathcal{G} := \{g : \mathbb{R}^d \rightarrow \{-1, 1\}\}$ be the set of classifiers suited to this setting. The classification error is defined as the map $L : g \in \mathcal{G} \mapsto \mathbb{P}(Y \neq g(X)) \in [0, 1]$, and we write $L^* := \inf_{g \in \mathcal{G}} L(g)$.

Let $\mathcal{F} := \{f : \mathbb{R}^d \rightarrow \mathbb{R}\}$ and $\text{sgn} : a \in \mathbb{R} \mapsto 2 \mathbb{1}_{\{a \geq 0\}} - 1 \in \{-1, 1\}$. In this exercise we focus specifically on the set of classifiers $\mathcal{G}_{\mathcal{F}} := \{\text{sgn} \circ f : f \in \mathcal{F}\}$.

Finally, let $\phi : \mathbb{R} \rightarrow \mathbb{R}_+$ be a differentiable, strictly convex, nondecreasing function satisfying $\lim_{x \rightarrow -\infty} \phi(x) = 0$ and $\phi(0) = 1$. For every $f \in \mathcal{F}$ (and hence every classifier in $\mathcal{G}_{\mathcal{F}}$) we consider the convexified version of the classification risk :

$$A(f) := \mathbb{E} \left[\phi(-Yf(X)) \right].$$

- 1) For every $u \in [0, 1]$ define $h_u : a \in \mathbb{R} \mapsto u \phi(-a) + (1 - u) \phi(a)$. Show that $\min_{f \in \mathcal{F}} A(f)$ is attained at

$$f^* : x \in \mathbb{R}^d \mapsto \arg \min_{a \in \mathbb{R}} h_{\eta(x)}(a). \quad (1)$$

Is this function well defined for every $x \in \mathbb{R}^d$?

- 2) Differentiate the function $h_{\eta(x)}$ for an arbitrary $x \in \mathbb{R}^d$, then check that $\text{sgn} \circ f^*$ coincides with the Bayes classifier.
- 3) (Zhang's lemma) Let $H : u \in]0, 1[\mapsto \min_{a \in \mathbb{R}} h_u(a)$ and assume there exist constants $s > 1$ and $c > 0$ such that

$$\forall u \in]0, 1[, \quad \left| \frac{1}{2} - u \right|^s \leq c^s (1 - H(u)). \quad (2)$$

- a) Let $f \in \mathcal{F}$. Show that $L(\text{sgn} \circ f) - L^* \leq 2 \left(\mathbb{E} \left[|\eta(X) - 1/2|^s \mathbb{1}_{\{(2\eta(X)-1)f(X) \leq 0\}} \right] \right)^{1/s}$.

- b) Let $A^* := \inf_{f \in \mathcal{F}} A(f)$. Deduce from the previous question that for every function $f \in \mathcal{F}$,

$$L(\text{sgn} \circ f) - L^* \leq 2c (A(f) - A^*)^{1/s}. \quad (3)$$

- c) What is H when $\phi = \exp$? For which constants s and c does condition (2) hold?